DEPARTMENT OF
APPLIED IT

An Investigation Into the Interplay of Physiological Responses, Personality Traits, and Subjective Discomfort During Simulated Driving Scenarios

Magnus Wahlström

Philip Andersson

Thesis:	15 hp
Program:	Cognitive Science
Level:	First cycle
Year:	2023
Supervisor:	Robert Lowe & Bahram Salamat Ravandi
Examiner:	Martyna Galazka
Report nr:	2023:049

Abstract

The emergence of autonomous vehicles has presented a myriad of challenges in the realm of human-computer interaction, particularly concerning autonomous braking systems. While technological advancements have streamlined these systems' safety and efficiency, the driver's interaction, characterized by physiological responses and personality traits, remains relatively understudied. This study investigates the nuanced relationship between drivers' galvanic skin response (GSR), personality traits based on the Big-Five model, and their subjective discomfort with autonomous braking systems in a semi-autonomous driving scenario.

The results revealed a significant positive correlation between the intensity of discomforting stimuli and both the physiological (measured by GSR) and self-reported discomfort levels. This suggests that an increase in stimuli levels generally corresponded to an increase in self-reported and physiological discomfort. Further, while personality traits such as agreeableness and openness to experience correlated negatively with self-reported discomfort levels, they showed no significant correlation with physiological responses. This indicates that while personality traits may influence the subjective experience of discomfort, they do not significantly impact the physiological response to discomforting stimuli.

This study contributes to the ongoing discourse on human-vehicle interaction design in the era of autonomous driving. It highlights the potential of utilizing objective physiological data to improve the subjective driving experience, thus laying the foundation for the development of personalized, adaptive braking systems. The findings also indicate that further research is necessary to understand better the relationship between personality traits and physiological responses to discomforting stimuli.

Keywords

Galvanic Skin Response (GSR), Big Five Personality Traits, BFI-10, Subjective Discomfort, Objective data, Automatic braking systems, Simulator

Sammanfattning

Framstegen med autonoma fordon har presenterat en rad nya utmaningar, bland annat när det gäller människans interaktion med dem. Tekniska framsteg har effektiviserat säkerheten och automatiseringen i dessa system. Men, vad detta har för implikationer för föraren när manuell kontroll över fordon minskar och automatiseringen ökas, är ett ämne som behöver undersökas ytterligare. Denna studie undersöker om det finns förhållanden mellan förarnas fysiologiska och psykologiska upplevelser och responser, med personlighetstyper som studeras genom bland annat Big Five-modellen, tillsammans med deras självrapporterade obehag vid uppvisad stimuli. Resultaten visade en signifikant positiv korrelation mellan ökning av obehag i stimuli, och ökad fysiologisk respons och självrapporterat obehag hos experimentdeltagare.

Denna studie bidrar till den pågående diskursen om utveckling av människa-maskin interaktion. Resultaten antyder potentialen att använda objektiva fysiologiska data för att förbättra den subjektiva körupplevelsen, samt justera individanpassade inbromsningssystem vid utveckling av nya bilar. Resultaten indikerar dock också att ytterligare forskning är nödvändig för att fullt ut förstå förhållandet mellan personlighetsdrag och fysiologiska svar på obehagliga stimuli.

Nyckelord

Galvanisk Hudreaktion (GSR), Big Five-personlighetsdrag, BFI-10, Subjektiv Upplevelse av obehag, Objektiv Data, Autonoma bromssystem, Simulator

Foreword

The study was conducted at Gothenburg University for the authors' bachelor thesis in cognitive science, in cooperation with Chalmers and Smart Eye. We want to extend huge thanks to Robert Lowe, Bahram Salamat Ravandi, Martin Bergström and William Hedley Thompson at Gothenburg University and Jonas Bärgman at Chalmers. Their insights, support and equipment during this work was crucial and much appreciated.

Table of Content

1 Introduction **7**
1.1 Background 7
1.2 A Gap in Knowledge 8
1.3 Research Questions and Strategy 9
1.3.1 GSR as a Proxy for Subjective Self Reporting 9
1.3.2 Increasing Precision of Predictability 9
1.3.3 Strategy 9

2 Previous Research **10**
2.1 GSR and Subjective Discomfort 10
2.2 Personality Traits and Physiological Response 10
2.3 Application in Automotive Safety 10
2.4 Summary 11

3 Method **12**
3.1 Methods 12
3.2 Participants 12
3.3 Procedure 13
3.4 Data Collection - Metrics and Instruments 16
3.4.1 Software and Data Processing 16
3.4.2 Galvanic Skin Response (GSR) 16
3.4.3 Big Five Personality Test 16
3.5 Data Processing 17
3.6 Statistical Analysis 17
3.7 Data Storage 18
3.8 Pilot Study 18

4 Results **20**
4.1 Validation of Stimuli Levels Through Self Reported Discomfort 20
4.2 GSR Validation - Correlation with Self Reported Discomfort 20
4.3 Correlational Analysis of BFI-10 Scores, GSR & Self Reported Discomfort 21
4.3.1 Correlation Between GSR Values and Self Reported Discomfort 21
4.3.2 Correlation Between Personality Trait Scores and GSR Values 21
4.3.3 Correlation Between Personality Trait Scores and Self Reported Discomfort 21
4.4 Concluding Remarks 22

5 Discussion **23**
5.1 Interpretation of Findings 23
5.2 Implications for Understanding Physiological Responses to Discomfort 23
5.3 The Role of Personality Traits in Experiencing Discomfort 24
5.4 Limitations 24
5.5 Ethical Considerations 25
5.6 Future Research, Practical Applications & Planned Further Analysis 25

6 Conclusion **27**

7 References **28**

8 Appendices **30**
Appendix 1 - Stimuli Validation, SPSS 30
Appendix 2 - GSR Correlation with Self Reports, SPSS 32
Appendix 3 - BFI-10 Correlation with GSR & Self Reports, SPSS 33
Appendix 4 - Required Sample Size Calculation 35
Appendix 5 - BFI-10 Table & Conversion Rules 36
Appendix 6 - iMotions Software 37
Appendix 7 - Experiment Slides: Instructions, Self Reporting & Countdown Screen 38

1 Introduction

1.1 Background

The advent of autonomous vehicles, propelled by groundbreaking advancements in automotive technology, has reshaped our perception of transportation. Although this progression is exciting, it necessitates a fresh understanding of human-computer interaction in the context of such sophisticated vehicles. This study is dedicated to investigating these dynamics, particularly concentrating on the driver's physiological responses, personality traits, and their intersection with the vehicle's autonomous systems. An extended aim is to utilize this data to fine-tune the automatic braking systems in vehicles, thereby elevating the driving experience.

Previous research in the realm of autonomous vehicles has mainly focused on refining technical algorithms and sensor systems to bolster the efficiency and safety of these vehicles. Still, there is an apparent research gap concerning the interplay between a driver's physiological reactions, personality traits, and interactions with autonomous driving systems, specifically automatic braking systems. We decided to step into this research gap, inspired by several studies that have touched upon different aspects of autonomous vehicles and their implications for traffic safety.

For instance, Nilsson et al. (2022) aimed to shed light on the impact of cognitive load on traffic safety, while Dillen et al. (2020) explored drivers' discomfort and anxiety when completely relinquishing control to an autonomous vehicle. Tavakoli et al. (2022) centered their research on understanding how drivers' behavior, anxiety, and emotional states could compromise traffic safety. Mostowfi et al. (2022) went a step further and studied biometric data obtained from eye metrics and facial expressions to decode specific emotions and behaviors. Moreover, Du et al. (2020) highlighted the problem of driver disengagement in fully autonomous vehicles. The studies above influenced the choice of experiment design for this study, both deciding on what to research - subjective experience of discomfort when presented with different traffic stimuli - as well as influencing the choice of methodological structure.

Common to the above studies is the use of certain data sensors to gather biometric data. Popular tools include Galvanic Skin Response (GSR) sensors that measure arousal by tracking sweat gland activity and eye-trackers that monitor eye movement in real-time. These tools, being consistent in their usage across studies, form a part of the research methodology of this study as well. Some studies also employed heart rate and facial detection measurements, with the latter providing insight into overall facial/head movement, like

retracting the head during a scare. To supplement these objective measurements, subjective self-reporting forms have also been used to gauge participants' discomfort levels.

Although there are researched models for predicting subjective states using a combination of objective measures, such as processing GSR, respiration, heart rate and ECG data through Support Vector Machines, as per Sharma & Gedeon (2012). These models have however focused mainly on predicting stress, not discomfort specifically. This study intends to build upon these foundations and open up a new avenue of research, in combining data from several objective sources, to increase precision of predictability of subjective discomfort. An increase in precision could for example prove vital for both traffic safety and user experience, in the ongoing development of automatic emergency braking systems in the automotive industry. The term “discomfort” is used in this study to represent the participants' own subjective experience of discomfort, instead of a more defined, and possibly complex, term such as “stress”. Because self reporting is used as a variable, “discomfort” was deemed general enough of a concept to produce comparable and relative subjective reporting, while “stress” could mean different things to different participants.

1.2 A Gap in Knowledge

The automatic braking system is an important safety feature in autonomous vehicles. However, its operation can lead to discomfort in some drivers, especially when the system is too sensitive and leads to frequent "false activations". The intersection of the vehicle and the driver is an important one, where safety is paramount but the driver's comfort and experience cannot be ignored. Hence, understanding the physiological and psychological dimensions of this discomfort is critical. Our study aims to probe whether objective data like physiological responses can be employed to predict drivers' subjective discomfort. This can then be applicable in the development of automatic emergency braking systems in the automotive industry, and elsewhere.

Employing a mixed-method approach, this study involved a semi-autonomous driving simulator to simulate real-world driving scenarios. Physiological data as well as personality trait information was gathered from twenty five participants. The objective was to identify correlations, from which individual user profiles could be formed in an extension to this study, and use these profiles to establish adaptive boundary conditions for automatic braking systems. These conditions could then potentially be adjusted in real-time according to the driver's reactions, enhancing the overall driving experience and potentially increasing safety.

The conclusions drawn from this study could contribute to the ongoing discussion around human-vehicle interaction design. They also highlight the potential of harnessing objective

data to improve the subjective driving experience, paving the way for the development of personalized, adaptive emergency braking systems, and beyond.

1.3 Research Questions and Strategy

The extended goal of the researchers is to use objective measures to effectively predict subjective states. This study aims to shed light on two main questions, detailed in the following sections.

1.3.1 GSR as a Proxy for Subjective Self Reporting

Can Galvanic Skin Response measures be used as a proxy to self reported discomfort? This is the first main question this study will explore. Objective proxies for self reporting of subjective states is crucial for applications in the automotive industry and in health care, where continuous physiological data can be collected, but self reporting information could be impractical or impossible.

1.3.2 Increasing Precision of Predictability

Can precision in predicting subjective states be increased by using Big Five personality scores? This is the second main question this study aims to shed light on. All objective proxies for subjective states can be said to be more or less crude approximations. Therefore it is vital to manage and be able to increase the precision of these approximations. This study will analyze in what way Big Five scores correlate with both GSR and self reporting data.

1.3.3 Strategy

The first level of analysis will validate the stimuli used in the experiments, in terms of correlation between both GSR data and self reported data. Five levels of discomfort-inducing stimuli will be used, and validation of these levels is a crucial prerequisite for the following analyses. The second level of analysis aims to explore the correlation between GSR data and self reported level of discomfort, to answer the first main question of this study. In the final level of analysis the second main question will be in focus, by exploring correlations between Big Five personality scores, GSR and self reporting data. By answering these two questions, this study hopes to lay ground for future research into further increasing precision of predictability, and to further increase the usability of objective proxies to subjective states.

2 Previous Research

The area of automotive safety, especially regarding autonomous or semi-autonomous vehicles, has been a growing focus in recent years. While much of the research has focused on the mechanics of the car itself, increasing attention is being given to the human component, namely the drivers' psychological and physiological states.

2.1 GSR and Subjective Discomfort

Numerous studies have explored the relationship between Galvanic Skin Response (GSR) and self-reported discomfort. The relationship between self-reported discomfort and physiological response is interesting to study since, as previous studies have shown, a higher level of reported subjective discomfort is related to increased GSR levels. For instance, Boucsein et al. (2012) provided a review of GSR as an index of emotional and cognitive states, demonstrating its applicability as a reliable measure of psychological stress. Moreover, a study by Sharma & Gedeon (2012) also highlighted the importance of GSR as an objective indicator of stress, correlating well with self-reported measures.

2.2 Personality Traits and Physiological Response

The Big Five personality traits—extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience—have been extensively researched in relation to various aspects of human behavior, cognition, and emotion. This personality framework has been shown to have significant influence on how individuals perceive, interpret, and react to stimuli in their environment, in a study by Naumann & Soto (2008). The correlation between some personality traits and stress response has been well established in the literature. Some research suggests that conscientious individuals might have better coping mechanisms and lower stress responses, for example Bibbey et al. (2013). Furthermore, extraversion has been associated with greater resilience to stress and less subjective discomfort, as per a study by Suls, Green, & Hillis (1998).

In terms of studies exploring the link between personality traits and GSR responses specifically, results have been mixed, with some studies suggesting that traits such as neuroticism may be associated with heightened GSR responses, for example Eysenck (1967). Others, like Figner & Murphy (2011), have found little to no relationship.

2.3 Application in Automotive Safety

The application of psychological and physiological data in the context of automotive safety is an emerging field. Some initial studies have already begun to explore this area. For instance, Healey & Picard (2005) used physiological measures, including GSR, to predict driver stress

in real-world situations. Similarly, a study by Dijksterhuis et al. (2011) demonstrated the potential for individual personality traits to predict driver behavior.

Bärgman et al. (2015) evaluated comfort/discomfort for the purpose of informing autonomous braking systems as to when to be deployed. However, this study focused on driver behavior rather than physiological variables.

While each of these domains has been investigated separately, the intersection of GSR, Big Five personality traits, and automotive safety represents a novel focus of research. The potential for individualized safety measures, where the car's automated systems can be calibrated based on the user's unique physiological and psychological responses, is a promising direction that could greatly enhance safety on the roads and driver comfort.

2.4 Summary

The role of agreeableness, conscientiousness, and openness to experience in discomfort and physiological responses has been less studied, indicating a need for further research.
Our study seeks to contribute to this field by examining both the self reported discomfort and physiological responses across five levels of discomfort-inducing stimulus, and investigating these responses with the Big Five personality traits.

3 Method

3.1 Methods

The current research was conducted with a mixed-methods approach involving quantitative data analysis. The following chapter describes the data collection process, information regarding participants, data collection instruments and data analysis process in detail.

3.2 Participants

A total of twenty seven adult participants were recruited for this study. The individuals were diverse in terms of age, gender, and driving experience to ensure a representative sample. Due to technical issues during data collection two participants' data was ignored, with an effective twenty five participants included in the final analysis. The participants were aged between 19 and 71, with a mean of 31,48. All had an active driver's license. 40% identified as female, 60% identified as male and no one identified as gender “Other”.

Recruitment was performed in a variety of ways. A form for registering as a participant was distributed online on various social media platforms, in school groups and elsewhere, as well as personally recruiting on campus and among friends and family. Printed posters were put up in various areas on campus in order to increase the participation further. To increase the level of participation, each participant was given a cinema ticket after the finished session.

3.3 Procedure

Figure 1. Example set-up of the driving “simulator”/rig.

The experiment was conducted in a controlled environment using a driving simulator. Each participant watched a series of ten video stimuli on a projector screen in a darkened room, each with a 10 second countdown, a 45 second driving video, followed by a self reporting screen on which participants were asked to report their level of discomfort. The session duration was approximately 30 minutes per participant, which included 10 minutes preparation and setup, 15 minutes of data collection, and 5 minutes for questions and comments after.

For the car simulator setup a metal frame was used with a car seat mounted, together with a computer steering wheel and pedal console. The GSR sensor is highly sensitive to movement, so to minimize any false readings, an arm rest was provided on which the participant was instructed to keep their left hand (with the attached GSR sensor) still during the session.

As participants arrived on their scheduled time slot, they were asked to take a seat in the simulator. They were given instructions on how to adjust the seat so that they sat comfortably and could reach the pedals. GSR-sensors were attached with velcro stripes to the inside of

their index- and middle finger of their left hand. After this they were given a brief instruction to emphasize key points, such as keeping their head and left hand as still as possible (for GSR stability). Before starting the session all sensors were controlled for proper function using the Sensor Preview monitoring-function of iMotions. Participants were then asked about their age, gender and if they had an active drivers license. All lights were then turned off in the room and participants were then given a last opportunity to ask questions before the session started.

The session started with a welcome screen followed by an instructions screen (Appendix 7, Figure 21). The instructions included information about their duties during the experiment, such as keeping their left hand rested on the support (for GSR stability), to mimic the behavior of the car on screen as closely as possible and to wait with any questions until after each video, but before proceeding beyond the self reporting screen. A computer mouse was provided for progression of the experiment, and the participants were asked to read all screens at their own pace, and click the "Next" button to continue. The participant was then given a trial run, preceded by a second instructions screen (Appendix 7, Figure 22, 23 & 24), with one group of stimuli containing a count down screen, a level 1 video and a self report screen. The count down screen was selected to be neutral in appearance, to allow for stabilizing of the participants sensor data after each stimuli. The ten driving videos were played in randomized order for each participant. The videos were divided into five levels of discomfortability with two videos of each level, and were recorded using a GoPro camera mounted on the dashboard inside a car with a viewpoint similar to that of the driver (forward facing with the front of the hood visible, but with no interior instrumentation in view).

Figure 2. Example of car setup when videoing scenarios that were used in the experiment.

The discomfortability levels were categorized by the following criteria:

Level 1 - No pedestrians visible.
Level 2 - Pedestrians visible but not in conflict with the cars path.
Level 3 - Pedestrians visible, but not in close proximity to the car. Potential of becoming a future conflict.
Level 4 - Pedestrians in potential conflict with the car's path and in closer proximity to the car, but not requiring drastic measures by the driver.
Level 5 - Pedestrians in potential conflict with the cars path and requiring drastic measures by the driver - hard turn and/or braking.

After each driving video, participants were presented with a self reporting screen, where they were asked to rate their peak level of discomfort during the previous video using a discrete scale between 1 to 6 (1 being low discomfort and 6 being high discomfort). These self-reported scores served as a subjective measure of discomfort for each stimulus.

After the ten trials of stimuli, participants were given a Big Five personality questionnaire which was presented on the screen and navigated with the mouse. The session ended with a “Thank You” screen where participants could choose to provide contact details in case they would want to receive a copy of the study after it was concluded, and contact details of the experiment leaders were provided. As lights in the room were turned on, participants were assisted in removing the GSR-sensors and to get out of the simulator. They were asked about their general experience during the experiment and about any feedback they had, although this data was not collected. They were then given a cinema ticket and assisted out of the building.

Participant immersion was an important factor to account for, since the experiment was done in a laboratory environment and since the participants had to simulate control of the car. To increase immersion a big projector screen close to the participant was used, the room was darkened during each session and noise canceling headphones were used for the participants. Furthermore, the steering wheel had force feedback enabled, which provided some resistance, simulating the feeling of turning a real steering wheel. Discussions were had regarding gamifying the tasks to increase immersion, but was discarded as it could introduce unforeseen factors affecting the results and sense of discomfort. Instead participants were instructed to mimic the car in the video using the controls, to draw focus to the stimuli. Participants were also asked to scan the environment in the video for potential conflicts, as they would in a real life driving scenario. These measures were determined to provide sufficient immersion, and this was later confirmed by participant feedback after the sessions. Graphically added turn signal indicator symbols were added to the videos to increase spatial awareness and planning.

The experiment leaders also moved to the back of the room in order to not distract the participant from the task at hand, and reduce the sense of being observed.

3.4 Data Collection - Metrics and Instruments

3.4.1 Software and Data Processing

iMotions version 9.3 was used on an OMEN laptop computer. iMotions was used to create the experiment and collect data from all the sensors. A slide show was created with information screens, stimuli trials and personality based questionnaires. The ten videos were each grouped with a respective self report screen, and this group was then randomized in order for each new participant. Later iMotions was used to do initial analysis of the results and to create annotations around our selected stimuli of interest. The relevant data was then exported into Microsoft Excel, in which structuring and cleaning of the data was done, before being imported into IBM SPSS Statistics (version: 29.0.0.0 (241)) for statistical analysis.

3.4.2 Galvanic Skin Response (GSR)

GSR measures skin conductivity, and can be used as an indicator of physiological stress. As stress levels increase, so does skin conductance. This is because of increased moisture on the skin surface, which increases electric flow. A GSR sensor measures the fluctuations in the flow of electricity, through varying electrical resistance, and the inversely varying conductance. This is measured by the use of two electrodes placed on the skin, in this study on the participants' left index- and middle finger. A small voltage is applied through the skin, so small it can not be perceived by a human, but is detectable after amplification. Changes in resistance (or conductivity) is measured between the two electrodes. Conductance can be measured anywhere on the skin, but is most effective on skin areas with a high density of eccrine/sweat glands. Fingers are commonly used for this reason, as well as being a relatively non-intrusive point of measurement. The output consists of registered values per time segment, which forms a varying curve that can be analyzed.

A Shimmer3 GSR detection system from iMotions was used for this study, with a sampling rate of 128 Hz. The data was sent from the electrodes to a receiving device connected to the iMotions software via Bluetooth for analysis and storage. GSR-metrics was used as an objective sign of autonomous physiological responses while experiencing discomfortable stimuli, by exporting unprocessed Conductance values from iMotions to Excel.

3.4.3 Big Five Personality Test

The Big Five personality test, or BFI (Big Five Inventory) was developed by John, Donahue & Kentle (1991). It measures a person's score on five different dimensions of personality traits, using a number of self rating questions. Originally forty four items, or questions, were

used. The personality trait dimensions consist of Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness.

For this study, a short version of the Big Five test was adopted, which includes ten items (Appendix 5, Figure 19) - two items per dimension. This short version, called BFI-10 (Big Five Inventory-10), was formed by Rammstedt & John (2007), to be applied when participant time is limited. Their study showed smaller effect sizes but still accurate correlations, as compared to the original or extended version of the BFI, which made the shorter model suitable for this study. The responses were submitted placing a marker along a scale between "Totally disagree" (to the left) and "Totally agree" (to the right). This was encoded as a value between 0 and 100. This was then converted using excel into a value between 1 and 5, where 0-19 = 1, 20-39 = 2, 40-59 = 3, 60-79 = 4 and 80-100 = 5. These were then converted into scores in each of the Big Five categories (Appendix 5, Figure 19).

3.5 Data Processing

To filter out irrelevant sensor data and avoid outliers each video was annotated in iMotions. The tag "Event" was added to all sections of the videos that contained elements from our level-categorization rules (Section 3.3, Procedure). This allowed for filtering out irrelevant data such as increase in GSR readings while turning outside of the event time frame.

The recorded study data was exported from iMotions and processed in Excel. GSR values for each participant at each time stamp during each video was listed in its own column. Then the average was calculated for each participant at each timestamp for each video, which gave an average GSR value per timestamp across all participants. These were then averaged with the GSR averages from the second level 1 video, which provided average values across all participants for both level 1 stimuli. Similar calculations were done for all 5 levels. By averaging the resulting GSR columns a single GSR value per level was obtained.

Using a similar method the average self reported discomfort levels was calculated for each level, which gave a single self report value for each level.

For correlational analysis between GSR, self report values and personality traits, average GSR and self report values per level for each participant was also calculated, which was then combined in Excel with each participant's Big Five scores.

3.6 Statistical Analysis

Analysis in SPSS was done in three steps. First a validation-step to verify that our chosen stimuli levels were indeed suitable for the study, in that they produced the intended step-wise

increase in self reported discomfort from level 1 to level 5. This was done with a One-Way Anova analysis, with a post hoc Bonferroni, as well as a test of homogeneity of variances using Levene's statistic. In a second validation-step a similar format of analysis was applied to the average GSR-data, to investigate whether it showed a similar step-wise increase in self reported discomfort. A Pearson's bivariate test was performed to discover correlations between each stimuli level, the average discomfort level, and the average GSR data for each level across all participants. As the third level of analysis a Pearson's bivariate test was performed to explore correlations between self reported discomfort, GSR-readings and Big Five personality trait scores.

Non-directional Pearson Product-Moment power analyses were calculated for each significant correlation discovered. This was because sample size in the study was relatively low, with twenty five participants. With a low sample size there is a potential risk that any significant results discovered could be inflated or unreliable, however this is covered more in detail in section 5.1, where the results are discussed.

Total number of participants in this study was $n = 25$, with ten stimuli at five levels of intensity, denoted by $N = 250$ instances. For certain analyses, such as the Tukey's B test, $n = 50$, representing twenty five participants reporting on two videos per each of the five stimuli levels.

3.7 Data Storage

The data was collected anonymously, with only the two variables gender and age given. The participants face's have been recorded during the experiment, but were not used in the analysis and will be destroyed after thesis submission. All related data was stored locally on a harddrive for the duration of the study.

3.8 Pilot Study

A pilot study was conducted in the weeks before public recruitment started. It was done with participants recruited among friends, family and student networks. In the early phases of the experiment, both in the planning phase and in the pilot phase, LED lights were used to light up the participant's face for the purpose of better uptake from the face tracking sensor. However, this was not used during the real data collection because of decreased sensor performance and reduced participant immersion, and the face tracking data was not included in the final analysis due to time limitations. Graphical turn signal indicators were also added to the driving videos after feedback, as not knowing which direction the car was going to turn increased confusion and reduced immersion. It was also decided to move all experiment leaders to the back of the room behind the participant, to reduce the risk of participants

feeling observed, which could again reduce immersion. Presentation of the Big Five questionnaire was also moved to after all video stimuli, since it could potentially affect participants' responses and behavior if they were asked about personality traits at the beginning of the session.

4 Results

4.1 Validation of Stimuli Levels Through Self Reported Discomfort

The primary objective of this study was to assess the correlation between five levels of stimuli and two distinct response measures: self-reported discomfort and GSR values. The five levels of stimuli were established, marked as 1 to 5, with Level 1 implying low discomfort and Level 5 implying high discomfort.

A one-way analysis of variance (ANOVA) was conducted to verify the validity of the chosen stimuli levels. The test included 250 cases. ANOVA results indicated a significant effect of stimuli level on the average self reported discomfort ratings, $F(4, 245) = 29.99$, $p < .001$. The effect size, denoted by η^2 .329, indicated that approximately 32.9% of the variance in self-reported discomfort can be explained by the level of stimuli.

The mean discomfort ratings progressively increased from Level 1 to Level 5, thereby validating the stimuli categorization used in this study. Specifically, Level 1 had the lowest self-reported average discomfort ($M = 1.42$, $SD = 0.702$) and Level 5 had the highest ($M = 3.54$, $SD = 1.343$). The corresponding results from the other levels are as follows; Level 2 ($M = 1.88$, $SD = 0.961$), Level 3 ($M = 2.06$, $SD = 0.935$), and Level 4 ($M = 2.80$, $SD = 1.309$).

A Bonferroni post hoc analysis was conducted to investigate pairwise comparisons of means of discomfort between the stimuli levels, at a 0.05 significance level. The results showed significant mean differences (Mdiff) between adjacent levels 3 and 4 (Mdiff = -0.740, $p = 0.007$) and levels 4 and 5 (Mdiff = -0.740, $p = 0.007$). However, there was no significant difference in discomfort reported between levels 1 and 2 (Mdiff = -0.460, $p = 0.339$) and 2 and 3 (Mdiff = -0.18, $p = 1.0$).

A test of homogeneity of variances using Levene's statistic was significant, $F(4, 245) = 8.049$, $p < .001$, indicating that the assumption of equal variances had not been met.

The low significance at lower levels of stimuli is expected, since levels 1 and 2 contained comparably lower arousing stimuli than levels 3, 4 and 5. In summary, these findings suggest a correspondence between stimuli levels and self-reported discomfort.

4.2 GSR Validation - Correlation with Self Reported Discomfort

A Pearson correlation analysis was performed to further validate the relationships between the Average GSR, Average self-report values and stimuli levels.

The Pearson correlation coefficient between average GSR and average self report value was found to be $r = 0.931$, $p < .021$, indicating a strong positive correlation. This suggests that as the average GSR levels increase, there is a correspondingly strong tendency for the average self report values to also increase.

Moreover, a highly significant correlation was found between the average GSR and the stimuli levels $r = .966$, $p < .008$ demonstrating a very strong positive relationship. This implies that as the levels increase, average GSR values tend to increase as well. The correlation between the average self report and the levels was significant $r = .979$, $p < .004$ suggesting a strong positive association. As the levels increase, there is a strong inclination for the average self report values to also increase.

These findings confirm that the lowest values for both GSR averages and self report averages would be observed at level 1 and the highest values at level 5. This analysis builds upon the previous results, providing additional statistical validation that increasing Levels are associated with increases in both physiological (GSR) and psychological (self reported) measurements.

4.3 Correlational Analysis of BFI-10 Scores, GSR & Self Reported Discomfort

This section reports the findings from the correlational analysis performed to investigate the relationships between personality trait scores (extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience), GSR levels, and self reported values.

4.3.1 Correlation Between GSR Values and Self Reported Discomfort

GSR for level 1 displayed a significant positive correlation with self-reported value at level 1 $r = .512$, $p = .009$, and a similar correlation was observed with GSR level 2 $r = .508$, $p = .009$.

4.3.2 Correlation Between Personality Trait Scores and GSR Values

No significant correlations were found between BFI-10 personality trait scores and average GSR values.

4.3.3 Correlation Between Personality Trait Scores and Self Reported Discomfort

Interestingly, there were significant negative correlations between agreeableness and self reported values at level 1 $r = -.492$, $p = .012$, and level 2 $r = -.435$, $p = .030$. In other words, higher agreeableness scores were associated with lower self reported values. Moreover, openness to experience displayed a significant negative correlation with self-reported values

at level 4 $r = -.426$, $p = .034$, indicating that individuals with higher openness tended to report lower values at this level.

4.4 Concluding Remarks

No significant correlations between personality traits and GSR responses could be observed. This suggests that while personality may influence the subjective experience of discomfort, it may not significantly impact the physiological response as measured by GSR. In summary, the results of this study revealed a complex interplay between personality traits, physiological responses, and self reported discomfort. It should however be noted that power analyses revealed either relatively high values, or low below 0.8, which means that any significance found should be viewed with caution.

5 Discussion

The findings from this study revealed some important insights about the correlation between discomfort-inducing stimuli, self-reported discomfort, and physiological response (measured by GSR), as well as their relationship with personality traits. The following chapter will contain discussions around the implications of our findings and their potential impact on future research and practical applications. The chapter will end with some future plans of exploring the data further, as well as data from other sensors collected during the experiments.

5.1 Interpretation of Findings

This study explored the complex relationship between drivers' physiological responses, measured via GSR, their Big Five personality traits, and the subjective experience of discomfort. The data collected revealed a significant correlation between average GSR and self-reported discomfort level across five varying traffic stimulus levels.

This correlation validates our initial hypothesis that GSR, an objective measure of physiological arousal, can be a useful indicator of subjective discomfort in drivers. These findings align with the previous research, Du et al. (2020) and Mostowfi et al. (2022), which highlighted the potential use of physiological data in assessing driver behavior and comfort.

Although significant results were discovered, power analyses revealed weaknesses. Most findings had power values below 0.8 (Appendix 3, Figures 13-17), which reduces reliability. Using a directional power test, rather than a nondirectional, some of those reached above 0.8, but nevertheless any significance should be seen with caution. Other findings had an associated power value of close to 1.0, which should similarly be seen as a potential decrease in reliability. Due to the small sample size these high power values could be inflated, reducing both reliability in the results and generalizability. The estimated sample size required for a power above 0.8 for $r = .426$ (the lowest power value that gave significant results), using a nondirectional test, is $n = 40$ (power = .800) (Appendix 4, Figure 18), which is well above the sample size of twenty five used in this study.

5.2 Implications for Understanding Physiological Responses to Discomfort

This study revealed a strong positive correlation between the intensity of discomfort-inducing stimuli and the physiological response as measured by GSR. This confirms the utility of GSR as a valid and reliable physiological measure of emotional arousal or discomfort. The strong positive correlation between average GSR and self-reported discomfort levels also suggests that participants' subjective experiences of discomfort corresponded closely with their

physiological responses. This concurrence between physiological and psychological responses can be used to validate the discomfort levels induced by different stimuli.

5.3 The Role of Personality Traits in Experiencing Discomfort

The results of the correlational analysis suggest that personality traits could influence the perception of discomfort. For instance, individuals who scored high in agreeableness and openness to experience reported lower discomfort levels. This might be because people with these personality traits tend to be more adaptable and tolerant of unpleasant stimuli.

However, personality traits did not significantly correlate with GSR responses, suggesting that these traits may not have a strong effect on the physiological responses to discomfort. This is an interesting finding because it suggests that while personality traits may influence how we perceive and report discomfort, they might not significantly alter our physiological reactions to discomforting stimuli.

5.4 Limitations

While this study offers valuable insights, it's worth acknowledging its limitations. A key constraint was the small sample size, leading to uncertain results, and the controlled laboratory setting, which might limit the generalizability of the results. Future research should consider larger, more diverse sample sizes, and a more advanced driving simulator or real-world driving scenarios for a more representative understanding. It is harder to achieve immersion if the driving is only fictitious, and variables such as the physical experience of movement - the feeling of acceleration, braking et cetera - are not present in a fixed simulator. However, placing experiment participants in a real-world driving situation with the purpose of exploring uncomfortable situations might present other problems, such as ethical ones and participant health risks. A more advanced, full motion simulator using virtual reality systems could be a more optimal setting, since it also allows for easier control of the laboratory environment.

Additionally, while analyses focused on GSR and personality traits, other physiological indicators (e.g. heart rate variability, eye tracking) and psychological factors (e.g. mood, stress levels) might also play a significant role in the experience of discomfort. We wanted to explore and use some of these variables for the analysis. However, due to limited time for this thesis, we had no possibility to follow through with this. We hope that further research considers these factors. There is also the possibility that some participants misunderstood the self reporting scale, and so reported a 6 (highest) instead of an intended 1 (lowest), since the scale was not labeled except for the numbers. There were however actions taken to minimize this risk, through written and oral instructions. Furthermore, despite the experiment leaders

moving to the back of the room during each session, participants could potentially feel observed and so give false self report responses, as well as responses to the personality questionnaire.

5.5 Ethical Considerations

The study was conducted in accordance with the ethical standards of the institutional and national research committee and with the 1964 Helsinki declaration and its later amendments. The participants gave informed consent in conjunction with choosing to proceed with the experiment.

Ethical considerations in regards to the experiment were made in the planning phase. The videos recorded were carefully planned out with the goal of not breaching the ethical guidelines or risk the health and safety of any pedestrian or participant in the study. The levels for the driving scenarios were also chosen so as not to induce high discomfort or chock.

5.6 Future Research, Practical Applications & Planned Further Analysis

Our findings suggest some avenues for future research. Firstly, further studies could explore other psychological and physiological variables that might influence discomfort responses, such as mood, stress levels, or heart rate variability. In addition, our study's results could also have practical implications for fields where understanding the human response to discomfort is crucial. For instance, in the healthcare industry, understanding patient discomfort can lead to improved pain management strategies and patient care procedures.

Data from a multitude of sensors was collected during the study. The planned use of these is further correlational analyses - to find novel combinations of sensor information, and to further increase the precision of objective data when predicting subjective states. Data was collected using EyeTracker (SmartEye, AI-X), face tracking software through a camera (Affectiva, incorporated into iMotions), motion of the steering wheel and pedals, as well as a Risk Propensity Score-questionnaire. The data collected includes pupil size and behavior, gaze velocity and acceleration, heartbeat information, and a host of emotive states based on facial expressions. The data can be explored using a machine learning model to identify the optimal combination of sensor data to predict subjective discomfort accurately. A Generative Adversarial Network (GAN) could then be used to create a synthetic database of plausible data instances, which could then be used to train another algorithm to produce and continually adjust user profiles for the industry. As real life data is being collected in live scenarios, these could replace the generated data for better applicability. Had there been time, analysis of gaze movement (a marked decrease of velocity and acceleration around the

intended events) and pupil distance from the sensor (an increased distance around the events, plausibly due to the wish to move away from potential danger) would have been included in this report, as an initial analysis showed promising correlations.

6 Conclusion

This investigation into the intricate relationships among discomforting stimuli, physiological responses (as measured by GSR), self-reported discomfort, and Big Five personality traits revealed significant and meaningful correlations. This understanding can have wide-ranging implications for fields where discomfort and its management are integral aspects, such as in the automotive or healthcare industry. While offering a foundational understanding, the practical application of this knowledge across industries could lead to the development of improved safety measures, pain management strategies, and overall enhanced well-being. As such, we anticipate our findings to contribute to future research and practical applications in areas where understanding human discomfort is vital.

The analyses demonstrated a strong correlation between the intensity of the discomforting stimuli and the physiological response as evidenced by the GSR readings. Moreover, a similarly strong correlation was noted between average GSR readings and self-reported discomfort levels. This supports the notion that physiological and psychological reactions to discomforting stimuli align closely, thereby verifying the effectiveness of our stimuli categorization. When considering the influence of personality traits, the results were fascinating. Although it was found that traits such as agreeableness and openness to experience correlated negatively with self-reported discomfort levels, these traits displayed no significant correlation with GSR readings. This suggests that while personality traits may influence the subjective experience and perception of discomfort, they seem not to significantly modulate the physiological response to discomforting stimuli.

In regards to the main research questions of the study, it can be concluded that GSR indeed can be used as a crude proxy to self reported discomfort. However it will need increased generalizability and accuracy. The second main question was intended to explore how this can be accomplished. Although correlations were found between some personality dimensions and self reporting, this could not be linked to GSR readings. These results point to the complexity of the interactions among discomfort-inducing stimuli, subjective discomfort, physiological responses, and personality traits. It underscores the necessity for further research to provide a more comprehensive understanding of these relationships. Specifically, future studies could focus on the exploration of other psychological and physiological variables such as eye-tracker readings for example gaze velocity and pupillometry, or heart rate variability.

In conclusion, the journey to comprehensively understand the human response to discomfort is complex and multi-faceted. However, it is believed that this study has contributed with insights to this quest, and we look forward to following future research and developments in this domain.

7 References

Bibbey, A., Carroll, D., Roseboom, T. J., Phillips, A. C., & de Rooij, S. R. (2013). Personality and physiological reactions to acute psychological stress. International journal of psychophysiology, 90(1), 28-36. doi: 10.1016/j.ijpsycho.2012.10.018

Boucsein, W., Fowles, D. C., Grimnes, S., Ben-Shakhar, G., Roth, W. T., Dawson, M. E., & Filion, D. L. (2012). Publication recommendations for electrodermal measurements. Psychophysiology, 49(8), 1017-1034. doi: 10.1111/j.1469-8986.2012.01384.x

Bärgman, J., Smith, K., & Werneke, J. (2015). Quantifying drivers' comfort-zone and dread-zone boundaries in left turn across path/opposite direction (LTAP/OD) scenarios. *Transportation Research Part F: Traffic Psychology and Behaviour*, *35*, 170-184. doi: https://doi.org/10.1016/j.trf.2015.10.003

Dijksterhuis, C., Brookhuis, K. A., & De Waard, D. (2011). Effects of steering demand on lane keeping behaviour, self-reports, and physiology. A simulator study. Accident Analysis & Prevention, 43(3), 1074-1081. doi: https://doi.org/10.1016/j.aap.2010.12.014

Du, N., Zhou, F., Pulver, E. M., Tilbury, D. M., Robert, L. P., Pradhan, A. K., & Yang, X. J. (2020). Predicting driver takeover performance in conditionally automated driving. Accident Analysis & Prevention. Advance online publication. doi: https://doi.org/10.1016/j.aap.2020.105748

Dillen, N., Ilievski, M., Law, E., Nacke, L. E., Czarnecki, K., & Schneider, O. (2020). Keep Calm and Ride Along: Passenger Comfort and Anxiety as Physiological Responses to Autonomous Driving Styles. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (pp. 1-12). ACM. doi: https://dl.acm.org/doi/10.1145/3313831.3376247

Eysenck, H. J. (1967). The biological basis of personality. Springfield, IL: Charles C. Thomas. https://www.scirp.org/(S(351jmbntvnsjt1aadkposzje))/reference/ReferencesPapers.aspx?ReferenceID=380149

Figner, B., & Murphy, R. O. (2011). Using skin conductance in judgment and decision making research. A handbook of process tracing methods for decision research, 163-184. https://www.researchgate.net/publication/228888313_Using_skin_conductance_in_judgment_and_decision_making_research

Healey, J., & Picard, R. (2005). Detecting stress during real-world driving tasks using physiological sensors. IEEE Transactions on intelligent transportation systems, 6(2), 156-166. https://affect.media.mit.edu/pdfs/05.healey-picard.pdf

John, O. P., Naumann, L. P., & Soto, C. J. (2008). Paradigm shift to the integrative big-five trait taxonomy. Handbook of personality: Theory and research, 114-158. https://www.researchgate.net/publication/289963274_Paradigm_shift_to_the_integrative_big_five_trait_taxonomy_History_measurement_and_conceptual_issues

Mostowfi, S., & Kim, J. H. (2022). Understanding Drivers' Physiological Responses in Different Road Conditions. In H. Krömker (Ed.), HCII 2022, LNCS 13335 (pp. 218–230). Springer Nature Switzerland AG. doi: 10.1007/978-3-031-04987-3_15

Nilsson, E. J., Bärgman, J., Ljung Aust, M., Matthews, G., & Svanberg, B. (2022). Let Complexity Bring Clarity: A Multidimensional Assessment of Cognitive Load Using Physiological Measures. Frontiers in Neuroergonomics, 3, 787295. doi: 10.3389/fnrgo.2022.787295

Rammstedt, B., & John, O.P. (2007). Measuring personality in one minute or less: A 10-item short version of the Big Five Inventory in English and German. Journal of Research in Personality, 41, 203–212. doi: 10.1016/j.jrp.2006.02.001

Sharma, N., & Gedeon, T. (2012). Objective measures, sensors and computational techniques for stress recognition and classification: A survey. Computer methods and programs in biomedicine, 108(3), 1287-1301. doi: 10.1016/j.cmpb.2012.07.003

Suls, J., Green, P., & Hillis, S. (1998). Emotional reactivity to everyday problems, affective inertia, and neuroticism. Personality and Social Psychology Bulletin, 24(2), 127-136. doi: 10.1177/0146167298242002

Tavakoli, A., Lai, N., Balai, V., & Heydarian, A. (2022). How are Drivers' Stress Levels and Emotions Associated with the Driving Context? A Realistic Study. arXiv preprint arXiv:2205.06116v2.

8 Appendices

Appendix 1 - Stimuli Validation, SPSS

Figure 1 - ANOVA - Self Reported Discomfort / Stimuli Level Group

ANOVA

Discomfort

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	139,400	4	34,850	29,990	<,001
Within Groups	284,700	245	1,162		
Total	424,100	249			

Figure 2 - Eta Squared - Self Reported Discomfort / Stimuli Level Group

ANOVA Effect Sizes[a]

		Point Estimate	95% Confidence Interval Lower	Upper
Discomfort	Eta-squared	,329	,228	,404
	Epsilon-squared	,318	,216	,394
	Omega-squared Fixed-effect	,317	,215	,393
	Omega-squared Random-effect	,104	,064	,139

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Figure 4 - Tukey's B - Stimuli Levels / Self Reported Discomfort

Discomfort

	Stimulus_Level	N	Subset for alpha = 0.05 1	2	3	4
Tukey B[a]	1	50	1,42			
	2	50	1,88	1,88		
	3	50		2,06		
	4	50			2,80	
	5	50				3,54

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 50,000.

Figure 5 - Bonferroni - Stimuli Levels / Self Reported Discomfort

Multiple Comparisons

Dependent Variable: Discomfort

	(I) Stimulus_Level	(J) Stimulus_Level	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
Bonferroni	1	2	-,460	,216	,339	-1,07	,15
		3	-,640*	,216	,033	-1,25	-,03
		4	-1,380*	,216	<,001	-1,99	-,77
		5	-2,120*	,216	<,001	-2,73	-1,51
	2	1	,460	,216	,339	-,15	1,07
		3	-,180	,216	1,000	-,79	,43
		4	-,920*	,216	<,001	-1,53	-,31
		5	-1,660*	,216	<,001	-2,27	-1,05
	3	1	,640*	,216	,033	,03	1,25
		2	,180	,216	1,000	-,43	,79
		4	-,740*	,216	,007	-1,35	-,13
		5	-1,480*	,216	<,001	-2,09	-,87
	4	1	1,380*	,216	<,001	,77	1,99
		2	,920*	,216	<,001	,31	1,53
		3	,740*	,216	,007	,13	1,35
		5	-,740*	,216	,007	-1,35	-,13
	5	1	2,120*	,216	<,001	1,51	2,73
		2	1,660*	,216	<,001	1,05	2,27
		3	1,480*	,216	<,001	,87	2,09
		4	,740*	,216	,007	,13	1,35

*. The mean difference is significant at the 0.05 level.

Figure 6 - Levene - Stimuli Levels / Self Reported Discomfort

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Discomfort	Based on Mean	8,049	4	245	<,001
	Based on Median	7,931	4	245	<,001
	Based on Median and with adjusted df	7,931	4	221,802	<,001
	Based on trimmed mean	8,470	4	245	<,001

Figure 7 - Mean Analysis of Stimuli Levels / Self Reported Discomfort

Report

Discomfort

Stimulus_Level	Mean	N	Std. Deviation
1	1,42	50	,702
2	1,88	50	,961
3	2,06	50	,935
4	2,80	50	1,309
5	3,54	50	1,343
Total	2,34	250	1,305

Appendix 2 - GSR Correlation with Self Reports, SPSS

Figure 8 - GSR Averages / Self Reported Discomfort / Stimuli Levels

Correlations

		GSR_Levels	Discomfort_Levels	Stimuli_Levels
GSR_Levels	Pearson Correlation	1	,931*	,966**
	Sig. (2-tailed)		,021	,008
	N	5	5	5
Discomfort_Levels	Pearson Correlation	,931*	1	,979**
	Sig. (2-tailed)	,021		,004
	N	5	5	5
Stimuli_Levels	Pearson Correlation	,966**	,979**	1
	Sig. (2-tailed)	,008	,004	
	N	5	5	5

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Figure 9 - GSR Averages / Self Reported Discomfort

Power Analysis Table

		Test Assumptions			
	Power[b]	N	Null	Alternative	Sig.
Pearson Correlation[a]	,767	5	0	,931	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Figure 10 - Stimuli Levels / GSR Averages

Power Analysis Table

		Test Assumptions			
	Power[b]	N	Null	Alternative	Sig.
Pearson Correlation[a]	,909	5	0	,966	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Figure 11 - Stimuli Levels / Self Reported Discomfort

Power Analysis Table

		Test Assumptions			
	Power[b]	N	Null	Alternative	Sig.
Pearson Correlation[a]	,959	5	0	,979	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Appendix 3 - BFI-10 Correlation with GSR & Self Reports, SPSS

Figure 12 - Pearson Bivariate Correlation

Correlations

		Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness	GSR_Level_1	GSR_Level_2	GSR_Level_3	GSR_Level_4	GSR_Level_5	Self_Report_Level_1	Self_Report_Level_2	Self_Report_Level_3	Self_Report_Level_4	Self_Report_Level_5
Extraversion	Pearson Correlation	1	,275	-,322	-,503*	-,378	,078	,065	,072	,092	,065	-,181	,033	,058	,054	,033
	Sig. (2-tailed)		,184	,116	,010	,062	,711	,758	,732	,661	,756	,386	,877	,783	,796	,876
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Agreeableness	Pearson Correlation	,275	1	,076	-,348	,405*	-,257	-,231	-,230	-,230	-,229	-,492*	-,435*	-,343	-,325	-,295
	Sig. (2-tailed)	,184		,718	,088	,045	,215	,267	,269	,269	,270	,012	,030	,093	,112	,153
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Conscientiousness	Pearson Correlation	-,322	,076	1	,104	,214	-,373	-,348	-,350	-,359	-,344	-,221	-,203	-,096	,116	,182
	Sig. (2-tailed)	,116	,718		,622	,305	,066	,089	,087	,078	,092	,288	,331	,649	,581	,385
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Neuroticism	Pearson Correlation	-,503*	-,348	,104	1	-,017	-,028	-,070	-,012	-,070	-,069	,132	,080	,035	,318	,204
	Sig. (2-tailed)	,010	,088	,622		,935	,896	,738	,955	,739	,743	,529	,704	,869	,121	,328
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Openness	Pearson Correlation	-,378	,405*	,214	-,017	1	-,204	-,167	-,167	-,189	-,167	-,216	-,390	-,176	-,426*	-,345
	Sig. (2-tailed)	,062	,045	,305	,935		,329	,425	,424	,364	,424	,299	,054	,399	,034	,091
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
GSR_Level_1	Pearson Correlation	,078	-,257	-,373	-,028	-,204	1	,990**	,990**	,992**	,993**	,512**	,268	,380	,044	,203
	Sig. (2-tailed)	,711	,215	,066	,896	,329		<,001	<,001	<,001	<,001	,009	,196	,061	,835	,330
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
GSR_Level_2	Pearson Correlation	,065	-,231	-,348	-,070	-,167	,990**	1	,985**	,992**	,990**	,508**	,239	,382	,016	,183
	Sig. (2-tailed)	,758	,267	,089	,738	,425	<,001		<,001	<,001	<,001	,009	,249	,060	,938	,383
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
GSR_Level_3	Pearson Correlation	,072	-,230	-,350	-,012	-,167	,990**	,985**	1	,989**	,992**	,456*	,215	,330	,019	,170
	Sig. (2-tailed)	,732	,269	,087	,955	,424	<,001	<,001		<,001	<,001	,022	,302	,107	,930	,415
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
GSR_Level_4	Pearson Correlation	,092	-,230	-,359	-,070	-,189	,992**	,992**	,989**	1	,995**	,509**	,220	,353	,033	,186
	Sig. (2-tailed)	,661	,269	,078	,739	,364	<,001	<,001	<,001		<,001	,009	,291	,084	,876	,374
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
GSR_Level_5	Pearson Correlation	,065	-,229	-,344	-,069	-,167	,993**	,990**	,992**	,995**	1	,490*	,214	,329	,022	,183
	Sig. (2-tailed)	,756	,270	,092	,743	,424	<,001	<,001	<,001	<,001		,013	,303	,109	,916	,381
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Self_Report_Level_1	Pearson Correlation	-,181	-,492*	-,221	,132	-,216	,512**	,508**	,456*	,509**	,490*	1	,643**	,695**	,504*	,576**
	Sig. (2-tailed)	,386	,012	,288	,529	,299	,009	,009	,022	,009	,013		<,001	<,001	,010	,003
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Self_Report_Level_2	Pearson Correlation	,033	-,435*	-,203	,080	-,390	,268	,239	,215	,220	,214	,643**	1	,642**	,468*	,525**
	Sig. (2-tailed)	,877	,030	,331	,704	,054	,196	,249	,302	,291	,303	<,001		<,001	,018	,007
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Self_Report_Level_3	Pearson Correlation	,058	-,343	-,096	,035	-,176	,380	,382	,330	,353	,329	,695**	,642**	1	,565**	,665**
	Sig. (2-tailed)	,783	,093	,649	,869	,399	,061	,060	,107	,084	,109	<,001	<,001		,003	<,001
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Self_Report_Level_4	Pearson Correlation	,054	-,325	,116	,318	-,426*	,044	,016	,019	,033	,022	,504*	,468*	,565**	1	,907**
	Sig. (2-tailed)	,796	,112	,581	,121	,034	,835	,938	,930	,876	,916	,010	,018	,003		<,001
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Self_Report_Level_5	Pearson Correlation	,033	-,295	,182	,204	-,345	,203	,183	,170	,186	,183	,576**	,525**	,665**	,907**	1
	Sig. (2-tailed)	,876	,153	,385	,328	,091	,330	,383	,415	,374	,381	,003	,007	<,001	<,001	
	N	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Figure 13 - GSR / Discomfort

Power Analysis Table

	Power[b]	Test Assumptions N	Null	Alternative	Sig.
Pearson Correlation[a]	,773	25	0	,512	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Figure 14 - GSR / Discomfort

Power Analysis Table

	Power[b]	Test Assumptions N	Null	Alternative	Sig.
Pearson Correlation[a]	,765	25	0	,508	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Figure 15 - BFI-10 Agreeableness / Discomfort

Power Analysis Table

		Test Assumptions			
	Power[b]	N	Null	Alternative	Sig.
Pearson Correlation[a]	,732	25	0	,492	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Figure 16 - BFI-10 Agreeableness / Discomfort

Power Analysis Table

		Test Assumptions			
	Power[b]	N	Null	Alternative	Sig.
Pearson Correlation[a]	,607	25	0	,435	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Figure 17 - BFI-10 Openness / Discomfort

Power Analysis Table

		Test Assumptions			
	Power[b]	N	Null	Alternative	Sig.
Pearson Correlation[a]	,587	25	0	,426	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Appendix 4 - Required Sample Size Calculation

Figure 18

Power Analysis Table

	N	Actual Power[b]	Test Assumptions			
			Power	Null	Alternative	Sig.
Pearson Correlation[a]	40	,800	,8	0	,426	,05

a. Two-sided test.

b. Based on Fisher's z-transformation and normal approximation with bias adjustment.

Appendix 5 - BFI-10 Table & Conversion Rules

Figure 19

A Brief Version of the Big Five Personality Inventory.

Big Five Inventory-10 (BFI-10)

Adapted from Rammstedt, B. & John, O. P. (2007). Measuring personality in one minute or less: A 10 item short version of the Big Five Inventory in English and German. *Journal of Research in Personality, 41*, 203-212.

Instructions: How well do the following statements describe your personality?

I see myself as someone who …	Disagree strongly	Disagree a little	Neither agree nor disagree	Agree a little	Agree strongly
1. … is reserved	(1)	(2)	(3)	(4)	(5)
2. … is generally trusting	(1)	(2)	(3)	(4)	(5)
3. … tends to be lazy	(1)	(2)	(3)	(4)	(5)
4. … is relaxed, handles stress well	(1)	(2)	(3)	(4)	(5)
5. … has few artistic interests	(1)	(2)	(3)	(4)	(5)
6. … is outgoing, sociable	(1)	(2)	(3)	(4)	(5)
7. … tends to find fault with others	(1)	(2)	(3)	(4)	(5)
8. … does a thorough job	(1)	(2)	(3)	(4)	(5)
9. … gets nervous easily	(1)	(2)	(3)	(4)	(5)
10. … has an active imagination	(1)	(2)	(3)	(4)	(5)

Scoring the BFI-10 scales (R = item is reverse-scored):

Extraversion: 1R, 5

Agreeableness: 2, 7R

Conscientiousness: 3R, 8

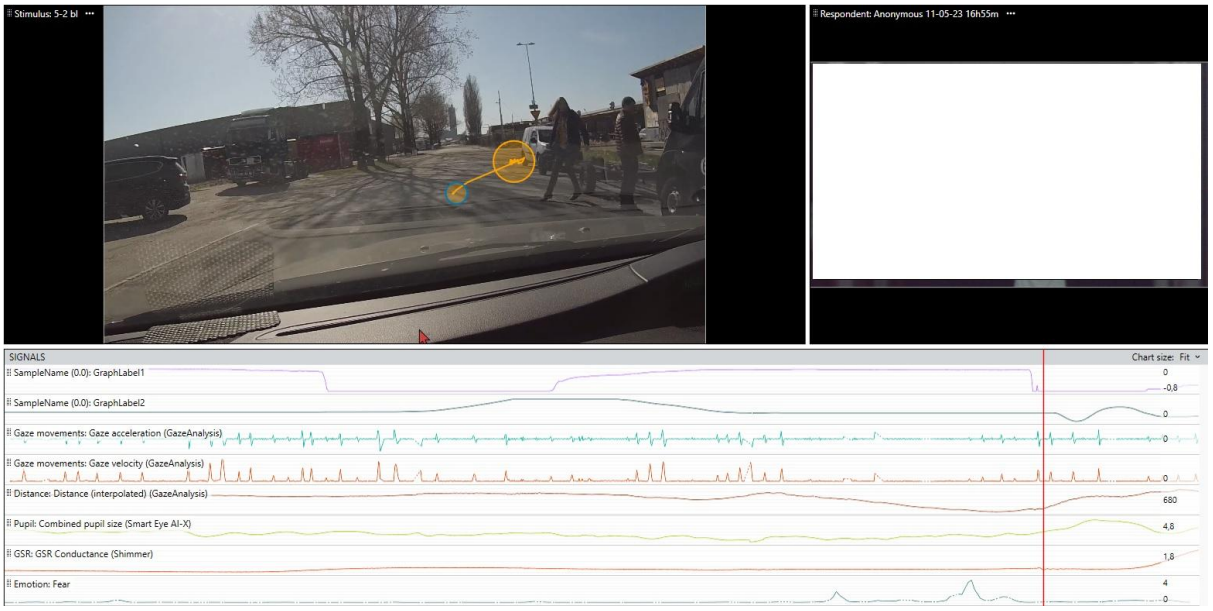
Neuroticism: 4R, 9

Openness to Experience: 5R, 10

(Retrieved 7/31/10 from http://www.ocf.berkeley.edu/~johnlab/pdfs/BFI-10.doc)

Appendix 6 - iMotions Software

Figure 20



Appendix 7 - Experiment Slides: Instructions, Self Reporting & Countdown Screen

Figure 21 - Initial Instructions Screen

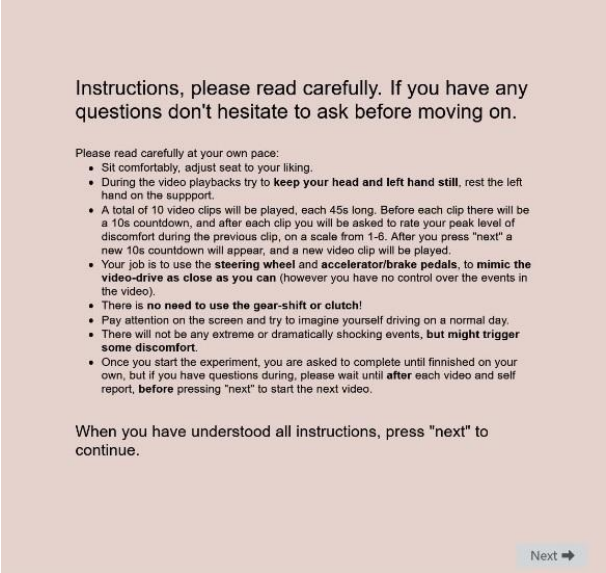


Figure 22 - Post Trial Run Instructions Screen

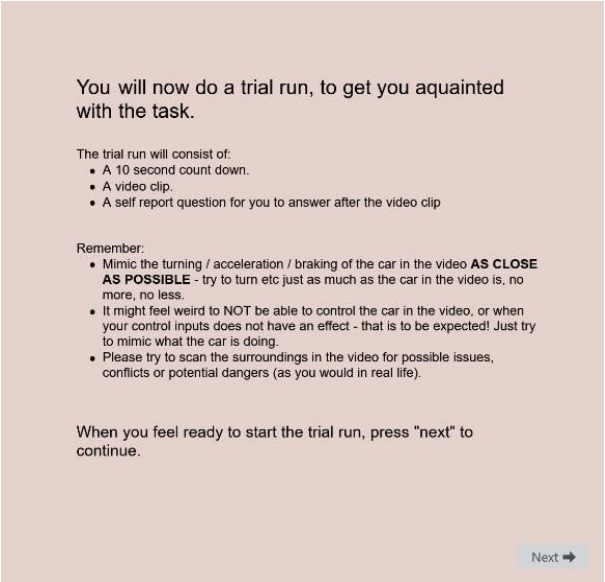


Figure 23 - Self Reporting Screen

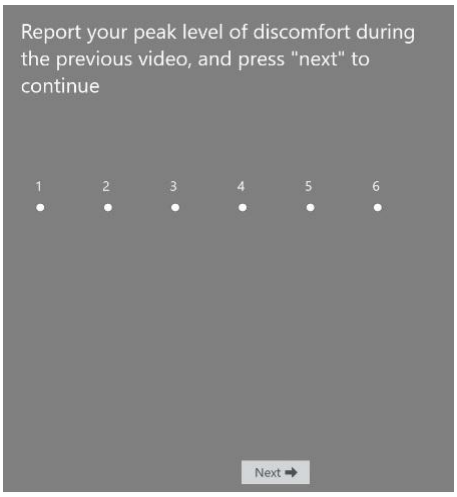


Figure 24 - Countdown Screen Example